

Last name _____

First name _____

LARSON—MATH 310—HOMEWORK WORKSHEET 05
Linear Independence and Gram-Schmidt

1. Write up a **neat** assignment on a **new sheet** of paper. (Do not cram your answers between the lines).
2. **Number** your problems so that it is easy to see what work matches the assigned problems.
3. Be verbose. Remember that you do not understand a concept if you do not know an **examples**.

Problems

1. Show that the list of vectors $\vec{a}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$ is linearly dependent.
2. We proved that any three vectors in $\mathbb{R}^2$ must be linearly dependent.
Show that $\vec{a}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ are linearly dependent.
3. Show that $\vec{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ are linearly independent.
4. Explain why $\vec{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ are a basis for $\mathbb{R}^3$.
5. Is the list of vectors $\vec{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ orthogonal? Explain.
6. Is the list of vectors $\vec{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$ orthogonal? Explain.
7. Is the list of vectors $\vec{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$ orthonormal? Explain.

(over)

8. (**Gram-Schmidt algorithm**). Input: $\vec{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\vec{a}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

Output: an orthonormal list of vectors $\vec{q}_1, \vec{q}_2, \vec{q}_3$.

(a) Let $\vec{q}_1 = \frac{1}{\|\vec{a}_1\|} \vec{a}_1$.

(b) Let $\tilde{q}_2 = \vec{a}_2 - (\vec{q}_1^T \vec{a}_2) \vec{q}_1$.

(c) Let $\vec{q}_2 = \frac{1}{\|\tilde{q}_2\|} \tilde{q}_2$.

(d) Let $\tilde{q}_3 = \vec{a}_3 - (\vec{q}_1^T \vec{a}_3) \vec{q}_1 - (\vec{q}_2^T \vec{a}_3) \vec{q}_2$.

(e) Let $\vec{q}_3 = \frac{1}{\|\tilde{q}_3\|} \tilde{q}_3$.

(f) Check that the list of vectors $\vec{q}_1, \vec{q}_2, \vec{q}_3$ is orthonormal.